BMC Microbiology

RESEARCH Open Access

Metagenomic report of element-microbe synergy and xenobiotic detoxification in the sacred waters of Khecheopalri lake, Eastern Himalaya

Rohit Das[1], Bipashna Thatal[1], Nagendra Thakur[1], Rakshak Kumar[2] and Buddhiman Tamang[1*]

Abstract

Background Khecheopalri Lake, a sacred freshwater body and recently recognized Ramsar Wetland site in Sikkim, India, holds both ecological and cultural significance. The ecological health of this lake is influenced by elemental inputs and environmental parameters, yet its microbial and functional diversity remain poorly characterized. In this study, we employed a multi-omics approach combining shotgun metagenomics, inductively coupled plasma mass spectrometry (ICP-MS), and culture-dependent analyses to provide an integrated understanding of the lake's microbial ecosystem. Shotgun metagenomics revealed taxonomic diversity and functional gene profiles, ICP-MS quantified elemental composition and its potential role in shaping microbial communities, while culture-dependent methods complemented metagenomic insights by isolating representative taxa. Together, these approaches highlight the interactions between microbes and elemental dynamics, offering new perspectives on the ecological functioning of this Himalayan wetland and its potential vulnerability to environmental change.

Results ICP-MS analysis revealed phosphorus (P) as the most abundant element, followed by iron (Fe), sodium (Na), magnesium (Mg), and potassium (K). Elevated BOD and COD levels in sample KES4 indicated organic pollution and coincided with the dominance of *Microcystis aeruginosa*, a cyanobacterium indicative of eutrophication. Shotgun metagenomic sequencing generated approximately 213 million reads, with bacteria constituting 98.85% of the community. Dominant phyla included *Pseudomonadota* and *Cyanobacteria*. Culturable isolates confirmed the presence of genera such as *Limnohabitans*, *Microcystis*, and *Mycolicibacterium*. Functional gene profiling showed that metabolism was the most enriched category (71.64%), with several genes (e.g., *xylB*, *pchF*, *clcD*) associated with xenobiotic degradation pathways.

Conclusion This first comprehensive metagenomic assessment of Khecheopalri Lake reveals diverse microbial populations involved in nutrient cycling and pollutant detoxification. The presence of genes linked to aromatic hydrocarbon degradation highlights the ecological potential of native microbes in mitigating environmental stress.

Keywords Culture dependent, Shotgun metagenomic, Sanger sequencing, Xenobiotic degradation

*Correspondence:
Buddhiman Tamang
bmtamang3@gmail.com

[1]Department of Microbiology, School of Life Sciences, Sikkim University, 6th Mile, Samdur, TadongGangtok 737102, Sikkim, India
[2]Department of Molecular Biology & Bioinformatics, Tripura University, Suryamaninagar, Tripura 799022, India

Background

Lakes serve as repositories for the inland carbon and nitrogen cycling because they absorb nutrients from nearby terrestrial ecosystems. Specialized assemblages of microorganisms can find a variety of habitats in lakes, such as the benthic layer, the chemocline and anoxic realms, the air–water interface, or the *aufwwuchs* (periphyton) on littoral microphytes [3]. Microorganisms in these water bodies carry out a large portion of biogeochemical cycling, breaking down complex organic matter and primary production [1]. This also imparts self-cleansing property of the freshwater lake where heterotrophic microorganisms decompose the organic matter, drives nutrient cycling, while autotrophic microorganisms fixes CO_2 and releases O_2 [7, 8]. There is always a balance between the nutrient availability and the nutrient uptake by the microorganisms in such ecosystems. Changes in microbial communities, mainly caused by nutrient levels, affect the water quality and the lake ecosystem [9].

Khecheopalri Lake, also known as *Khecheopalri Pemachen Tsho* or *Tsho-Shu Tsho*, located at an altitude of 1,700 m in the West District of the Sikkim in the Eastern Himalaya (27°22′24" N, 88°12′30" E) is a sacred freshwater lake [14]. According to Buddhist belief, Guru Padmasambhava, the patron saint of Sikkim once preached to sixty-four *dakinis*, the female embodiment of enlightenment, at the lake. As believed by the Bhutia-Lepcha Buddhists, the lake's dakini endows the water with the power to purify households during and after death ceremonies [14]. Because of its "wish fulfilling" belief, it is the most venerated of Sikkim's 150 lakes, having significant spiritual significance and acting as a popular pilgrimage and tourism site [45]. The lake spans 3.79 hectares with an average depth of 7.2 m, lies within the 12 km^2 forested Ramam watershed, surrounded by broadleaved forest. Situated on the southern boundary of the Kangchendzonga Biosphere Reserve (Buffer Zone IV), the lake is fed by two perennial and five seasonal inlets and is drained through internal seepage and an outlet stream. It was formed over 3,500 years ago during early Holocene, the lake is believed to have been sculpted by ancient glacial activity, with a moraine ridge marking the Lethang Valley's southern boundary [51].

Khecheopalri Lake has been the focus of biodiversity conservation efforts since the mid-1990s. Recently in July 2024, this lake has been included in the Ramsar Convention of Wetlands (https://www.ramsar.org/). Surveys have recorded 682 species from five kingdoms, including 460 plant species (311 genera, 126 families) and 165 bird species (119 genera, 49 families), with several classified as threatened by the IUCN. The lake is notable for studies on bacterial, chromista, and fungal diversity, identifying 30 chromist species, 20 fungi, and two bacteria [62]. Two major vegetative communities dominate the peatland: *Temperate Sphagnum Bog*, characterized by *Sphagnum nepalense*, *S. palustre*, and *Vaccinium* species, and Warm-temperate Moist Deciduous Forest. Algal, diatom, and phytoplankton surveys have reported over 60 species, with phytoplankton density ranging from 0.75×10^4 to 19.45×10^4 units L^{-1}. Zooplankton, including genera like *Daphnia*, *Cyclops*, and *Brachionus*, showed densities between 0.70×10^4 and 2.20×10^4 units L^{-1} [50]. The bog also supports 11 species of waterfowl, including the critically endangered Baer's Pochard *(Aythya baeri)*. Fish species identified include Grass Carp (*Ctenopharyngodon idella*) and Giant Danio (*Danio aequipinnatus*), with additional genera like *Garra* and *Schistura* [45].

Although the lake is conserved and its sacredness maintained by the local people, the disturbances in the catchment area may affect the lake biological and physiochemical aspects [22]. Given the lake's pristine yet culturally active setting, the microbial community structure is expected to reflect both the natural geochemical environment and localized environmental pressures [23]. In spite of its importance, the microbial biodiversity of Khecheopalri Lake is poorly understood, especially using high-resolution, culture-independent metagenomic approaches. Present information is mostly limited to flora and fauna information, which represents a minority of the community and ignores potentially important taxa participating in nutrient dynamics and pollution management. Microbial community profiling complemented with elemental and physicochemical analyses can be the indicator of physiochemical changes and the environmental stress [10].

This research proposes to describe the taxonomic and functional richness of microbial communities at Khecheopalri Lake through metagenomic shotgun sequencing, emphasizing the connection between community structure and local variations in nutrient and elemental compositions along the lake. By distinguishing unique microbial assemblages and putative bioindicators of pollution or environmental stress, the research hopes to reveal ecological functions and mechanisms of adaptation of the microorganisms in this high-altitude, sacred ecosystem. These results are anticipated to give significant information on the dynamics of nutrients, health of ecology, and resilience of holy freshwater systems and hence contribute to wider programs of environmental surveillance, conservation management planning, and sustainable aquatics management of Himalayan aquatic ecosystems.

Methodology

Sampling and site description

In the present study, water samples were collected from seven sites of Khecheopalri Lake, starting from the source (KES0) and gradually moving towards the anthropogenic areas of the lake (KES1–6) [22]. At each site, 100 mL of water was collected in triplicates as a composite sample by pooling water from three different points within ~ 5 m radius at a depth of ~ 1 m. The samples were collected in sterile flasks, sealed immediately under water to minimize aerial contamination, and transported in a 4 °C cooler. Samples were stored at –20 °C until further analysis (Fig. 1).

Elemental analysis

All elemental analysis was performed using Shimadzu ICPMS 2030 coupled with AS-10 autosampler. The instrument configuration and operation parameters were RF power (1.20 KW), sampling depth (5 mm), plasma gas (Argon 8.0 L/min), auxiliary gas (Argon 1.10 L/min), Carrier Gas (0.70 L/min), nebulizer (Nebulizer, 07UES), Chamber (Cyclone Chamber), Chamber Temperature (5 °C), No. of Scans (10 times), Cell Gas (He) (6.0 ml/min), Cell Voltage (–21.0 V), and Energy Filter (7.0 V). Water sample were collected in three independent replicate and pooled in equal quantity to form a total of 1 µl and was immediately mixed with 1% ultrapure HNO_3 and stored at 4°C for further analysis. The calibration standards for the analyzed elements were prepared in different concentration ranges depending on whether they were major or trace elements. For the major elements Li, Na, Mg, K, Ca, and P, the calibration range was 100–10,000 µg/L, while for the trace and toxic elements Mn, Fe, Co, Ni, Cu, Zn, As, Se, Mo, Ag, Cd, Pb, and Cr, the calibration range was 1–200 µg/L and was prepared using multi-element stock solutions from Merck and Sigma Aldrich were used for calibration standards preparation by serial dilution in 1% HNO_3 ultrapure water with resistivity > 18 mΩ·cm (Milli-Q) and ultra-pure nitric acid. Phosphorus in water samples was quantified using ICP-MS at m/z 31 with collision/reaction cell interference control. Samples for dissolved P were field-filtered (0.45 µm) and acidified to pH < 2 with ultrapure HNO_3, while total phosphorus was determined after persulfate digestion (121 °C, 30–45 min), followed by cooling, dilution to volume, and preservation with HNO_3. Calibration standards (100, 1000, 5000, 10,000 µg/L) were prepared from a certified phosphorus stock in 1% HNO_3 and spiked with an internal standard (45Sc, 10 µg/L). Measurements were performed in He KED mode at m/z 31 or in O_2 reaction mode with a mass shift to PO^+ at m/z 47, using three replicates per read and routine quality control checks including blanks, spikes, duplicates, and CRMs. Concentrations were blank-corrected, dilution-adjusted, and reported as µg P/L. Further one way ANOVA and Kruskal–Wallis test was performed to find the significance between samples using PAST4.0 software [17].

Physiochemical parameters of the sampling sites

In situ monitoring of major physico-chemical parameters, including conductivity, water temperature, dissolved oxygen (DO), salinity, pH, total dissolved solids (TDS), and turbidity, was carried out using a Multiparameter pH/ORP/EC/TDS/Salinity/DO/Pressure/Temperature Waterproof Meter (HI98194, Hanna Instruments) equipped with a 4 m cable. The probe was allowed to stabilise for approximately 30 s prior to measurement to ensure sensor equilibration. Further one way ANOVA and Kruskal–Wallis test was performed to find the significance between samples using PAST4.0 software [17].

Pollution index calculation

The Pollution Index (PI) was calculated for each element to evaluate the extent of contamination in Khecheopalri Lake. The single-factor PI was derived using the formula: $\mathbf{PI}_i = \mathbf{C}_i/\mathbf{S}_i$. where C_i is the measured concentration of

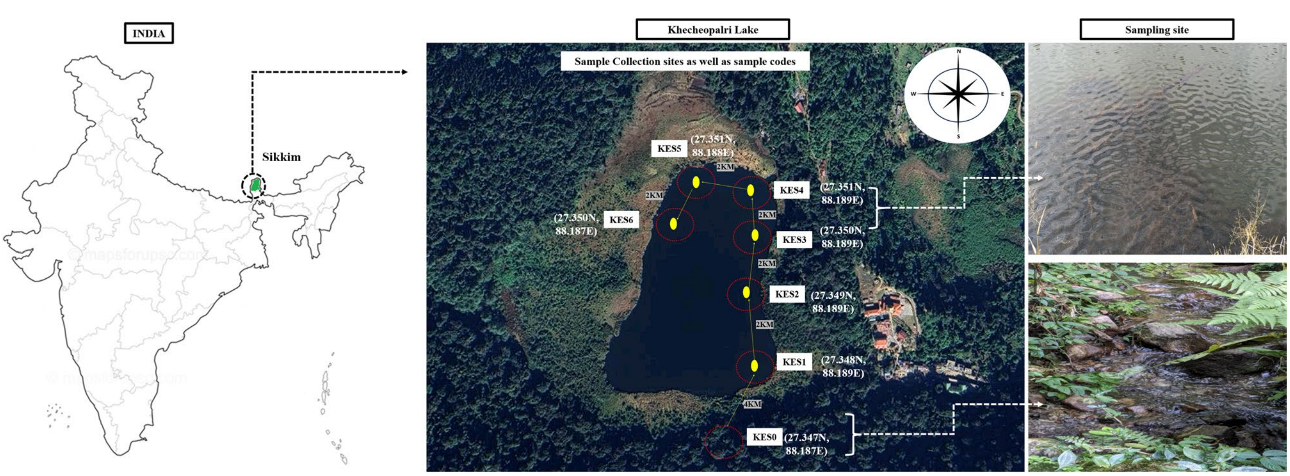

Fig. 1 Samples collected from 7 distinct sites of Khecheopalri lake and has been coded using prefix KES, the map was taken from google map earth view and precise latitude and longitude was recorded using GARMIN India eTrex® 32x

element i in water (ppm) and S_i is the corresponding permissible limit as per WHO (2017/2022) and BIS IS:10,500 (2012) standards. To evaluate the overall pollution level at each site, the Comprehensive Pollution Index (CPI) was computed as:

$$\mathrm{CPI} = \frac{1}{n}\sum_{i=1}^{n}\mathrm{PI}i$$

where n is the number of parameters considered. The classification of the Pollution Index (PI) was used to interpret the contamination level of each element, where $\mathrm{PI} \leq 1$ indicates concentrations within permissible limits, $1 < \mathrm{PI} \leq 2$ denotes slight pollution, $2 < \mathrm{PI} \leq 3$ reflects moderate pollution, and $\mathrm{PI} > 3$ represents high pollution; accordingly, the Comprehensive Pollution Index (CPI) was applied to assess the overall pollution status of each site, with CPI values greater than 1 considered indicative of pollution.

Biological oxygen demand and chemical oxygen demand calculation

BOD was determined following the 5-day incubation method. Water samples were collected in BOD bottles without air bubbles. Initial dissolved oxygen (DO) was measured using a DO meter (Hach HQ1130 Series, India). The samples were incubated at 20 °C for 5 days in the dark to prevent photosynthesis [20]. After incubation, the final DO was recorded. BOD was calculated using the formula:

$$\mathrm{BOD}\,(mg/L\,(ppm)) = DO_{initial} - DO_{final}.$$

COD was analyzed using the open reflux method. A 10 mL water sample was digested with 5 mL of potassium dichromate (0.25 N) and 15 mL of sulfuric acid containing silver sulfate. The mixture was refluxed for 2 hours at 150°C. After cooling, the excess dichromate was titrated against standard ferrous ammonium sulfate using ferroin indicator. COD was calculated as:

$$\mathrm{COD}\,(mg/L\,(ppm)) = (Vb - Vs) \times N \times 8000/Sample\ volume\,(ml)$$

where:

- V_b = Volume of FAS for blank (mL)
- V_s = Volume of FAS for sample (mL)
- N = Normality of FAS

Further one way ANOVA and Kruskal–Wallis test was performed to find the significance between samples using PAST4.0 software [17].

Culture-independent techniques

Metagenomic DNA extraction

All water samples were filtered prior to DNA extraction using a laboratory vacuum filtration (QIAvac 24 Plus vacuum manifold, QIAGEN) setup equipped with disposable filter funnels and sterile 0.22 µm membrane filters to capture microbial biomass. The membranes were aseptically cut and transferred into PowerWater Bead Pro Tubes (QIAGEN DNeasy PowerWater Kit) and processed according to the manufacturer's protocol. Briefly, 1 mL of lysis buffer (Solution PW1; PowerWater Lysis Solution, prewarmed to 55 °C) was added, and the tubes were vortexed at maximum speed for 5 min to achieve mechanical bead beating and chemical lysis, which fragmented the filter surface and released microbial DNA. The resulting lysate (~600 µL) was clarified by centrifugation, followed by inhibitor removal (Solution IRS; Inhibitor Removal Solution, 2–8 °C, 5 min), binding of nucleic acids to MB Spin Columns using PW3 (PowerWater Binding Solution), washing with PW4 (PowerWater Wash Solution) and ethanol, and final elution in 100 µL EB (Elution Buffer). When required, the eluted DNA was further concentrated by ethanol precipitation as recommended in the kit protocol. Environmental DNA (eDNA) was extracted using the XpressDNA Bacteria Kit (50) (MG20PIL-50) from MagGenome Technologies Pvt. Ltd., Chennai, India, following the manufacturer's protocol. DNA quality and concentration were measured using a Qubit Fluorometer (Thermo Fisher Scientific, USA), with a detection range of 10–100 ng/µL. Additionally, DNA integrity was assessed by performing 0.8% agarose gel electrophoresis.

Shotgun metagenomic sequencing

The eDNA samples underwent paired-end DNA sequencing (DNA-Seq) library preparation using the Illumina HiSeq 4000 platform, following the manufacturer's protocol. Libraries were constructed using the Illumina TruSeq DNA PCR-Free Library Preparation Kit (Illumina Inc., USA), with fragment sizes of ~350 bp. Illumina-specific adapters were ligated to both ends of the DNA fragments. In cases where low input DNA required amplification, a limited-cycle PCR (8 cycles) was performed to enrich adapter-ligated fragments and ensure sufficient library yield. Sequencing was carried out using a 101 bp paired-end module. Post-sequencing, quality control of the reads was conducted using the NGSQC Toolkit [48] to obtain high-quality (HQ) reads in FASTQ format. A stringent filtering criterion of 80:30 was applied, considering only reads with over 80% HQ bases and a Phred score ≥ 30 for genome assembly. The HQ paired-end libraries were trimmed using Trimmomatic v0.36 [4] with default parameters on the Galaxy platform. Finally, the trimmed reads were assembled into contigs using the de Bruijn graph-based assembler, MetaSPAdes [43].

Metagenomic data analysis and Functional pathway analysis

The classification was inferred by running Kaiju against the NCBI BLAST nr + euk database, using the greedy run mode parameter. The obtained results were further categorized into five taxonomic levels: domain, phylum, family, genus, and species using the script GOTTCHA [15]. Annotations, function and composition of bacterial communities were done using Prodigal, which predicts protein-coding genes [21], with ORFs shorter than 180 nucleotides excluded from further analysis by default. tRNA genes were identified using the ARAGORN program [30], while ribosomal RNA genes (5S, 5.8S, 16S, 18S, 23S, 28S) were detected and classified using rRNAFinder [13]. All ORFs were then searched against the Kyoto Encyclopedia of Genes and Genomes (KEGG) database [25] for KEGG ID annotation. Functional attributes were obtained at three levels: Level-1, Level-2, and Level-3, corresponding to the category, super pathway, and pathways, respectively [12]. To investigate genes associated with important pathways related to pollution degradation, genome annotation was first performed using Prokka (v1.14.6) [55] available on the Galaxy Australia platform (https://usegalaxy.org.au/). Assembled genome sequences in FASTA format were uploaded and annotated with default parameters. The gene were then mapped against KEGG pathway database (https://www.genome.jp/kegg/pathway.html).

Statistical analysis

The correlation of element, BOD, and COD was analysed using canonical correlation analysis (CCA) using PAST software [17]. The non-parametric Shannon index, Simpson's index of diversity (1-D), Chao-1, and evenness were computed using PAST software v.4.0 were also calculated with Paleontological Statistics Software Package for Education and Data Analysis (PAST) v4.0 [17]. Relationship between microorganisms in samples and various levels of functional profiles, represented by different super-pathways, was executed through non-parametric Spearman's rank correlation using IBM SPSS software v20 and PAST v4.0 following the description of Martino et., (2019) [37]. Correlation profiles were displayed via a heatmap develop [39] web tool [39]. The unique and shared species was analysed using R package (Rvenn). The bacterial differential and function pathway analysis to know the significant dissimilarity index between source (KES0) and highly anthropogenic active area (KES3 and KES4) using ANCOM-BC [36]. The dissimilarity was visualized using volcano plot.

Molecular characterization via culture dependent analysis

Isolation of bacterial isolates

The water samples were enriched in different enrichment media such as SOC Broth (M1379), R-2A Broth (M1687), Luria Bertani Broth, Miller (M1245), Nutrient Broth (M002), Plate Count Agar (M091A), and BG-11 broth (M1541), each enrichment broth was incubated at 37 °C for 24 h, except the BG-11 broth which was placed in a CO_2 incubator. After enrichment the pure colonies were isolated by spread plate and streak plate method [41, 42]. Isolated pure colonies were characterized based on their colony morphology [41, 42].

Identification of bacterial isolates and phylogeny

A total 109 bacterial isolates were obtained from seven water samples. Based on morphological characterization, 18 bacterial isolates were selected for partial 16S rRNA gene-based identification. Genomic DNA was extracted using the Qiagen QIAamp DNA Mini Kit (Qiagen, USA) following the manufacturer's protocol. The 16S rRNA gene was amplified according to the method described by [5], and the PCR products were analyzed via gel electrophoresis. PCR purification was performed using the QIAquick PCR Purification Kit (Qiagen, USA) before sequencing. The purified 16S rDNA was sequenced with the BigDye™ Terminator v3.1 Cycle Sequencing Kit (Applied Biosystems, USA) on an Automated DNA Sequencer (ABI Genetic Analyzer 3500). Sequence assembly was done manually using FinchTV 1.4, and taxonomic identification was performed through BLASTn analysis against the NCBI nr/nt database. Multiple sequence alignment with related genomes was carried out using Clustal Omega [61]. A phylogenetic tree was constructed using the Neighbor-Joining method [54] with Tamura–Nei evolutionary distance calculation [59] in MEGA 11 software. The phylogenetic tree was visualized and edited using the Interactive Tree of Life (iTOL) tool [32].

Results

Sample description and Physiochemical parameter analysis

Water samples collected from seven sites (KES0–KES6) across Khecheopalri Lake revealed a clear environmental gradient influenced by increasing anthropogenic activity. The source site (KES0) showed clearer water with low turbidity (2.0 NTU), lower total dissolved solids (TDS; 50 ppm), lower salinity (0.02 ppt), and comparatively higher dissolved oxygen (DO; 9.0 ppm). In contrast, downstream sites, particularly KES3 to KES6, recorded higher turbidity (5.0–13.0 NTU), higher TDS (110–220 ppm), elevated salinity (0.05–0.13 ppt), and reduced DO (4.8–7.5 ppm), together indicating an increase in organic load and anthropogenic influence along the gradient, a more detailed description of the results is provided below. Consistent sampling at ~ 1 m depth across a 2 km transects ensured spatial resolution of microbial and chemical changes. This gradient provided the framework for correlating microbial diversity and function with environmental impact.

The concentration of 19 elements in parts per billion (ppb) were evaluated through ICP-MS. Phosphorus (P) contains the highest concentration in all the samples, ranging from 3352.231 (KES1) to 4666.264 (KES4), followed by Iron (Fe), which ranges from 1029.268 (KES1) to 1856.399 (KES3). Sodium (Na), Magnesium (Mg), and Potassium (K) are also found in substantial quantities. Trace elements like Cobalt (Co), Nickel (Ni), Copper (Cu), Zinc (Zn), Arsenic (As), and Selenium (Se) occur in comparatively small amounts, whereas Molybdenum (Mo) and Silver (Ag) occur in very low or trace amounts in certain samples. KES4 contains the highest Phosphorus (4666.264) and Potassium (455.2773) among the samples, whereas KES3 contains the highest Iron (1856.399) and Manganese (36.73133) concentrations. Similarly, KES1 tends to have lower levels of the majority of elements than the other samples. The higher concentration elements are illustrated in Fig. 2a and Supplementary File 1: Table 1 and lower concentration are illustrated in Fig. 2b and Supplementary File 1: Table 1. The elemental analysis across samples revealed significant differences in group means and medians. One-way ANOVA showed a clear effect of sampling site on elemental concentrations, with between-group variation accounting for a substantial portion of the total variance ($F = 4.82$, $p = 0.0003$; permutation $p = 0.0002$). The variance components indicated a moderate intraclass correlation (ICC = 0.261) and a medium effect size ($\omega^2 = 0.20$). However, Levene's tests from both means ($p = 0.041$) and medians ($p = 0.036$) suggested heterogeneity of variances, which was further confirmed by a significant Welch's test ($F = 5.12$, $df = 58.3$, $p = 0.0007$). In agreement with the parametric analysis, the non-parametric Kruskal–Wallis test also indicated significant differences in elemental medians between groups ($H = 16.27$, χ^2, $df = 6$, $p = 0.012$), confirming that sample sites differ significantly in their elemental composition.

In situ measurements of the physico-chemical parameters of Khecheopalri Lake revealed distinct spatial variations across the sampling sites (Supplementary File 1: Table 2). The source point (KES0) exhibited the least polluted condition, with low conductivity (80 µS/cm), TDS (50 ppm), turbidity (2.0 NTU), and the highest dissolved oxygen (9.0 ppm), indicating relatively pristine water quality. A gradual increase in pollution-related parameters was observed from KES1 to KES2, with conductivity rising to 150 µS/cm, TDS to 90 ppm, and turbidity to 4.0 NTU, accompanied by a slight decline in DO (8.2 ppm).

The most deteriorated water quality was recorded at KES3 and KES4, where conductivity reached 350–360 µS/cm, TDS 210–220 ppm, and turbidity peaked at 12.5–13.0 NTU. Dissolved oxygen dropped markedly to 5.0 and 4.8 ppm, respectively, while pH values declined to 6.8 and 6.7, reflecting polluted and stressed conditions. Beyond these sites, a gradual improvement was evident, with KES5 and KES6 showing reduced conductivity (250 and 180 µS/cm), TDS (150 and 110 ppm), and turbidity (7.8 and 5.0 NTU), alongside partial recovery of DO levels (6.2 and 7.5 ppm). One-way ANOVA revealed highly significant differences in the physicochemical parameters across the sampling sites ($F = 20.27$, $p = 5.72 \times 10^{-11}$; permutation $p < 0.001$). Between-group variation accounted for a large proportion of the total variance, with a high intraclass correlation (ICC = 0.734) and a strong effect size ($\omega^2 = 0.702$). Variance component analysis indicated that site-related variability (Var(group) = 6,565.22) was substantially larger than residual error (Var(error) = 2,384.96). However, Levene's tests from both means ($p = 4.55 \times 10^{-11}$) and medians ($p = 1.12 \times 10^{-6}$) suggested heterogeneity of variances, which was supported by a significant Welch's test ($F = 1095$, $df = 16.14$, $p = 3.91 \times 10^{-20}$). The Kruskal–Wallis test further confirmed significant differences in physicochemical parameters across the sampling sites (H = 43.14,

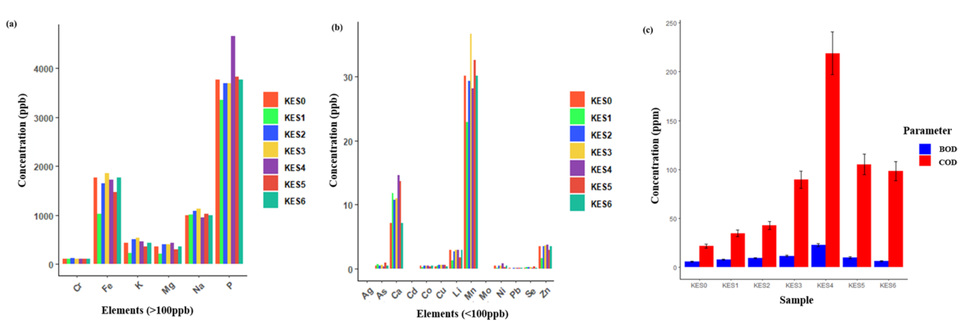

Fig. 2 **(a)** Elemental analysis of each individual site via ICPMS > 100 ppb. (**b**) Elemental analysis of each individual site via ICPMS < 100 ppb. (**c**) Biological and chemical oxygen demand analyzed across seven sites of Khecheopalri Lake

tie-corrected Hc = 43.15, df = 6, $p = 1.09 \times 10^{-7}$), indicating that the sample medians varied markedly between locations.

Most elements, including Fe, Mn, Pb, Cr, As, and Se, exceeded the permissible drinking water limits set by WHO/BIS, with KES4 showing the highest PI values for Fe (≈115), Mn (≈56), and Cr (≈223), along with the highest CPI, confirming severe contamination; this was further supported by elevated BOD (22.8 ppm) and COD (218.9 ppm), indicating significant organic pollution buildup. Sites KES3 and KES5 exhibited moderate to high CPI values, suggesting localized pollution stress, whereas KES0, KES1, and KES6 showed comparatively lower PI and CPI values, indicating better water quality. Overall, the Pollution Index (PLI) analysis classified the sites as follows: KES0 (0.85, baseline), KES1 (0.92, slight pollution), KES2 (1.05, low pollution), KES3 (1.85, high pollution), KES4 (2.10, severe pollution), KES5 (1.35, moderate pollution), and KES6 (1.15, low–moderate pollution).

The biochemical oxygen demand (BOD) and chemical oxygen demand (COD) concentrations were determined in seven samples (KES0–KES6). BOD concentrations ranged from 5.6 ppm (KES0) to 22.8 ppm (KES4), while COD concentrations ranged from 21.9 ppm (KES0) to 218.9 ppm (KES4). KES4 had the highest levels of pollution, reflecting severe organic contamination (Fig. 2c). The one-way ANOVA for BOD and COD across sampling sites revealed significant differences in mean values, with between-group variation explaining a substantial proportion of the total variance (F = 6.18, $p = 0.0322$; permutation p = 0.031). Variance component analysis showed that site effects contributed moderately to the overall variation (ICC = 0.463), with a moderate effect size ($\omega^2 = 0.302$). Tests of homogeneity indicated unequal variances (Levene's test for means: $p = 0.0259$; medians: $p = 0.0381$), and the Welch's test further confirmed significant differences between sites (F = 6.18, df = 5.07, $p = 0.045$). The Kruskal–Wallis test for BOD and COD showed a significant difference in sample medians across sites (H = 7.41, tie-corrected Hc = 7.41, df = 1, $p = 0.0065$), confirming that organic load and water quality varied significantly between the sampling locations.

Culture independent

Taxonomic classification and abundance profiling

A total of 213,628,884 reads were obtained from the samples with an average of 15,259,206 reads per sample. The total number of bases recovered from the samples averaged to be 2.3 Gbp with an average GC% of 50% (Supplementary File 2). Shotgun metagenomic sequence analysis of all the water samples showed different domains, viz., bacteria, archaea, viruses, plantae, protist and eukaryotes. The total relative abundance was calculated in all the samples where bacteria were the most abundant domain (98.85%), followed by eukaryota (0.78%), viruses (0.20%), archaea (0.068%), protist (0.054%) and plantae (0.022%) (Fig. 3a). Overall, 6 phyla, 30 families, 118 genera, and 4751 species were detected. Pseudomonadota was the most abundant phylum in samples KES0, 1, 2, 5 and 6 whereas cyanobacteria were highest in samples KES3 and 4 (Fig. 3b). At family level, Enterobacteriaceae was abundant for samples KES0, 1, 2, 5 and 6. Whereas, Microcystaceae was found in abundance in samples KES3 and 4 (Fig. 3c). At the genus level, *Limnohabitans* was found to be in high abundance in samples KES0, 1, 2, 5 and 6 (85.05%, 49.39%, 64.21%, 41.50%, and 44.24%). Whereas, *Microcystis* was found in abundance in samples KES3 and 4 (51.80% and 66.83%) (Fig. 3d). Differential abundance testing with ANCOM-BC revealed clear compositional shifts between source and anthropogenic groups. Several bacterial genera were significantly enriched in the source samples, including *Limnohabitans* and *Bradyrhizobium*, which showed high positive log2 fold changes (p < 0.05). In contrast, genera such as *Microcystis* and others were significantly depleted in the source group, indicating increased representation under human-impacted conditions (Fig. 3e). At species level, samples KES0, 1 and 2 had *Limnohabitans planktonicus* was found to be in highest abundance, in sample KES3 *Candidatus Planktophila dulcis* was found in an abundance of 13.04%, whereas in sample KES4 *Microcystis novacekii* was found in an abundance of 13.99%, in sample in KES5 *Limnohabitans planktonicus* and KES6 *Limnohabitans radicicola* was found in abundance respectively. The non-parametric Shannon, Simpson, and Evenness indices were highest in the KES5 (0.76 ± 0.06, 1.83 ± 0.13, and 0.52 ± 0.04), whereas, chao-1 was highest in KES1 (21 ± 1.5) (Fig. 4a and Supplementary File 1: Table 3).

Canonical correlation analysis at the genus level with BOD, COD, and elemental analysis revealed that an increase in *Microcystis, Mycobacterium, Klebsiella, Polynucleobacter*, and other genera showed a positive correlation with BOD and COD. This suggests that their presence could serve as an indicator of water pollution in aquatic ecosystems. Another Canonical Correlation Analysis (CCA) revealed a strong relationship between major elemental components and specific bacterial genera, including *Klebsiella, Pseudomonas, Caulobacter, Sphingomonas, Polaromonas, Novosphingobium, Bdellovibrio,* and *Sinobaca*. This suggests that these bacterial groups may be influenced by or play a role in the biogeochemical cycling of key elements in the environment. Their presence and abundance could be indicative of specific elemental compositions in the water, potentially reflecting natural geochemical processes or anthropogenic influences (Fig. 4b). Ag and Mo was highly correlated to *Microcystis, Mycobacterium, Bradyrhizobium* etc.

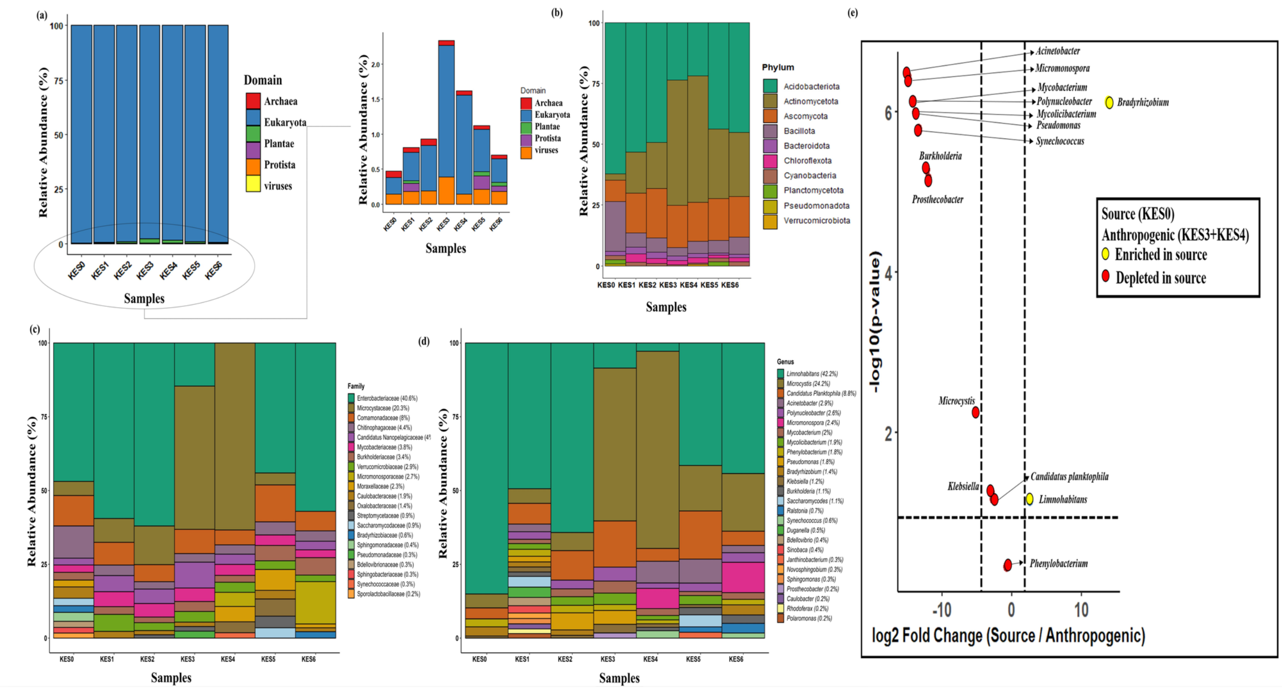

Fig. 3 Metataxonomic analysis from seven sites of Khecheopalri Lake represented in stacked bar plot of (**a**) domain > 1% and < 1% abundance. (**b**) phylum > 1% abundance (**c**) family > 1% abundance (**d**) genus > 1% abundance. (**e**) Differential abundance analysis using ANCOM-BC between source (KES0) and highly anthropogenic area (KES3 + KES4) visualized using volcano plot

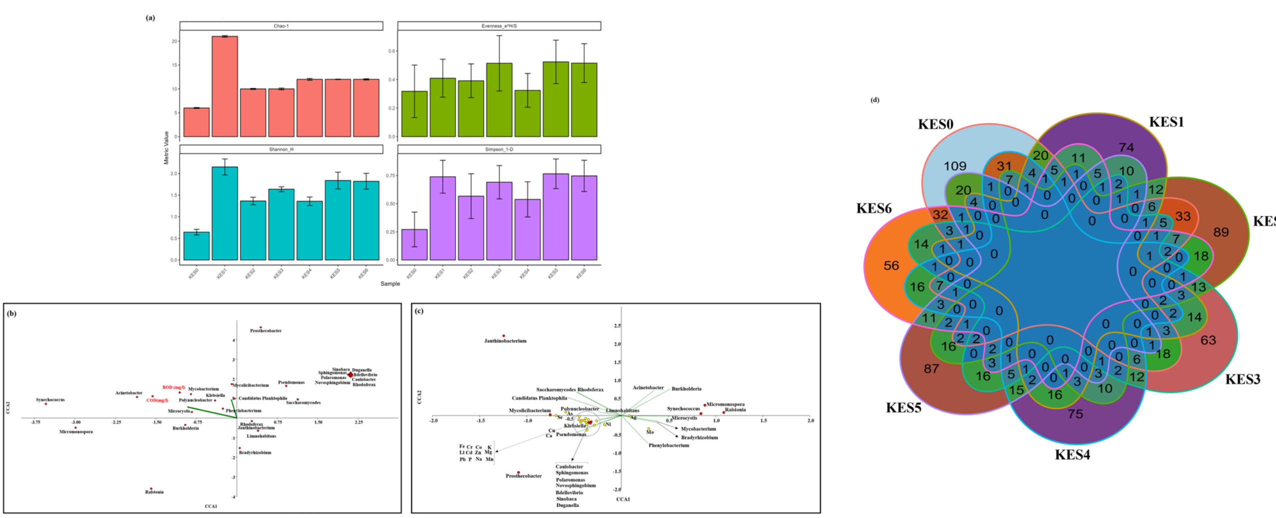

Fig. 4 (**a**) Alpha diversity indexes of the seven sites of Khecheopalri Lake. (**b**) Canonical correspondence analysis of Biological and chemical oxygen demand with > 1% genus abundance. (**c**) Canonical correspondence analysis of elements from seven sites of Khecheopalri Lake with > 1% genus abundance. (**d**) Shared and unique species from seven sites of Khecheopalri Lake

A cyanobacterial genus that produces toxic algal blooms, which may accumulate or interact with trace metals such as Ag and Mo, potentially by biosorption or enzymatic mechanisms. *Mycobacterium* is a genus that includes both environmental and pathogenic species, some of which can thrive in metal-enriched environments by potentially utilizing metals as cofactors in enzymatic processes. Additionally, it includes nitrogen-fixing bacteria that are typically associated with plant roots but can also be found in aquatic environments. Molybdenum is a requirement for nitrogenase, the enzyme that fixes nitrogen, and this could be the reason for its association with *Bradyrhizobium* (Fig. 4c). A diverse microbial community of both shared and unique species, KES0, 1, 2, 3, 4, 5, and 6 had unique species of 109, 74, 89, 63, 75, 87 and 56. The number of shared species 32, 75, 72, 25, 79, 45, and 30 respectively in the seven samples (Fig. 4d).

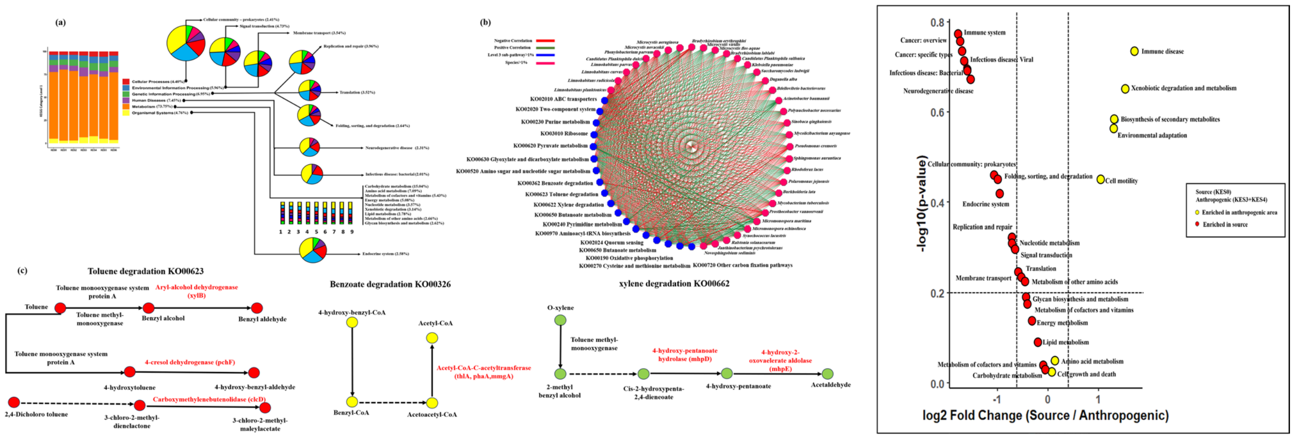

Fig. 5 (**a**) Predictive functionality of annoted KEGG pathway from category and super-pathway from seven sites of Khecheopalri Lake. (**b**) Positive and negative correlation of abundant genus and abundant sub-pathway from seven sites of Khecheopalri Lake (**c**) KEGG pathways related to xenobiotic degradation annotated via Prokka. (**d**) Differential pathway analysis using ANCOM-BC between source (KES0) and highly anthropogenic area (KES3 + KES4) visualized using volcano plot

Functional pathway analysis

Alignment of the metagenomic ORFs to KEGG databases yielded various functional pathways which were enriched next. The quantities of prodigal ORFs recovered from the samples were 17,456 (KES0), 40,872 (KES1), 31,456 (KES2), 25,267 (KES3), 19,812 (KES4), 43,228 (KES5), and 33,812 (KES6). About 72% of the total ORFs was mapped against KEGG functional pathway. The functional pathways were further classified into 3 subclasses using the KEGG mapping. In level 1, metabolism (71.64%) was found to be most abundant followed by genetic information processing (6.95%), human diseases (6.28%), environmental information processing (5.96%), organismal systems (4.75%), and cellular processes (4.40%). At level 2, carbohydrate metabolism (15.04%) was the most abundant followed by amino acid metabolism (7.09%), Metabolism of cofactors and vitamins (5.43%), Energy metabolism (5.08%), Nucleotide metabolism (3.57%), Xenobiotic degradation (3.14%), Lipid metabolism (2.78%) etc. as illustrated in Fig. 5a.

Spearman's rank correlation was done between some of the major microorganisms in all the water samples and different major Level 3 super-pathways obtained from KO annotation as illustrated via network analysis in Fig. 5b. It was observed that highest abundance of *Limnohabitans planktonicus* in sample KES0, 1 and 2 was positively purine metabolism (KO00230), *Candidatus Planktophila dulcis* in sample KES3 was positively to Quorum sensing (KO02024). *Microcystis aeruginosa,* abundant in KES4 was also mostly positively correlated to xylene degradation (KO00622) and lastly, *Limnohabitans radicicola* of KES6 was positively correlated to pyruvate metabolism (KO00620).

Genes associated with xenobiotic degradation pathways were identified in the analyzed microbial genomes, indicating the organisms' capacity to metabolize aromatic hydrocarbons such as toluene, xylene, and benzoate. For toluene degradation, key enzymes including aryl-alcohol dehydrogenase (*xylB*), 4-cresol dehydrogenase (*pchF*), and carboxymethylenebutenolidase (*clcD*) were annotated. In the case of benzoate degradation, enzymes such as acetyl-CoA acetyltransferase, encoded by *thlA*, *phaA*, and *mmgA*, were detected. Additionally, the xylene degradation pathway was supported by the presence of 2-keto-4-pentanoate hydrolase (*mhpD*) and 4-hydroxy-2-oxovalerate aldolase (*mhpE*). These findings highlight the metabolic versatility and potential ecological roles of the microbial community in xenobiotic transformation and environmental detoxification (Fig. 5c).

Functional annotation and differential pathway analysis indicated marked differences in metabolic potential between source and anthropogenic environments. Pathways related to xenobiotic biodegradation (log2FC = 1.5), immune disease (log2FC = 1.69), and environmental adaptation (log2FC = 1.3) were enriched in anthropogenic samples, reflecting microbial adaptation to stress and pollutant exposure. In contrast, core functions such as translation (log2FC = –0.54), replication and repair (log2FC = –0.74), and nucleotide metabolism (log2FC = –0.69) were relatively depleted, suggesting a reduction in basal cellular processes. Additionally, pathways associated with infectious diseases and cancer were overrepresented in anthropogenic sites, pointing toward shifts toward pathogenic and stress-related functional repertoires (Fig. 5d).

Molecular characterization via culture dependent analysis

Isolation, characterization, identification and phylogeny of bacterial isolates

A total of 109 isolates were isolated from 7 sampling site based on the colony morphology 18 isolates were selected for 16SrRNA analysis. 3 isolates from KES0 were selected having round colonies with approximately 1–2 mm colony size and unpigmented, after gram staining it was found to be gram-negative, and rod shaped, which was identified as *Limnohabitans planktonicus.* 3 isolates from KES1 were selected with 2 having circular, convex, unpigmented, smooth surface and gram negative and was identified as *Limnohabitans curvus,* while another one was identified as *Limnohabitans planktonicus.* 2 colonies were selected from KES2 sample one was approximately 1–2 mm colony size, unpigmented, rod shaped and gram-negative which was identified as *Limnohabitans parvus.* Whereas, second isolate had distinct yellow color, with small, smooth colony, gram positive and rod shaped, which was identified as *Mycolicibacterium anyangense.* The sample KES3 was majorly dominated by large, spherical colonies that grew into green sheet like over time, 3 colonies were isolated and was identified as *Microcystis aerugonisa.* Similar, cases were observed in KES4 sample 3 isolates were selected which also come to be *Microcystis aerugonisa.* 2 isolates were selected from KES5 which was identified as *Limnohabitans planktonicus.* Lastly, from KES6 we selected 2 isolates, 1 was identified to as *Limnohabitans curvus* and other had a small, orange colonies, cocci shaped and gram positive which was identified as *Micromonospora maritima.* Phylogenetic tree analysis was done using Tamura Nei model and the evolutionary history was inferred by Neighbor-joining method. The percentage of replicate tree in which the associated taxa clustered together in the bootstrap test (500 replicates) are shown next to the branches, all the identified bacteria were seen to form clade with its own type strains obtained from NCBI database as shown in Supplementary Fig. 1.

Discussion

The current research gives a detailed evaluation of physiochemical parameters, microbial diversity, and functional profiles of Khecheopalri Lake. The combination of elemental analysis with culture-independent microbiological techniques gives useful information about the lake's ecological status and possible environmental stressors [7, 8]. The elemental analysis also showed high levels of phosphorus (P) and iron (Fe) in all the water samples, with KES4 having the highest level of phosphorus (4666.264 ppb). Phosphorus is an essential nutrient that affects primary productivity in aquatic communities [19]. High phosphorus concentrations have been strongly linked to eutrophication and the proliferation of cyanobacterial blooms, as observed in Taihu Lake, China [35]. In a similar study, Read et al., [52] used ^{31}P NMR spectroscopy to characterize diverse phosphorus forms in Lake Mendota, including orthophosphate, monoesters, diesters, pyrophosphate, polyphosphate, and phosphonates, and demonstrated that their temporal variations correlated with phytoplankton biomass and water quality. Packa et al., [46] established a two-dimensional ion chromatography–electrospray ionization high-resolution mass spectrometry (2D-IC-ESI–MS) method for unbiased phosphate measurement in the Canadian Great Lakes, achieving low detection limits (5 μg L^{-1}) and enabling simultaneous analysis of phosphate-containing metabolites. Similarly, Reitzel et al., [53] applied ^{31}P NMR spectroscopy to sequential extracts from lake sediments to identify sedimentary phosphorus compounds, while Cantarero et al., [6] validated inductively coupled plasma mass spectrometry (ICP-MS) for both total and dissolved phosphorus in agricultural runoff, showing comparable results to traditional UV–Vis methods with high sensitivity (5 μg L^{-1}). A similar trend may be occurring in Khecheopalri Lake, where a high density of *Microcystis aeruginosa* was detected in samples KES3 and KES4.

This elevated microbial presence could be attributed, at least in part, to the practice of fish feeding by visitors, which introduces organic matter and nutrients particularly phosphorus into the water [63]. Such nutrient enrichment can disrupt the lake's ecological balance, promoting the growth of bloom-forming cyanobacteria [28]. Additionally, elevated iron levels, which play a crucial role in microbial metabolic processes such as nitrogen fixation and electron transport [16], may further support microbial proliferation. The combined impact of these elements suggests that anthropogenic nutrient input from fish feeding may be a contributing factor to the observed microbial shifts in the lake. High Fe levels in KES3 (1856.399 ppb) can affect the growth of iron-using microorganisms. Trace elements like cobalt (Co), nickel (Ni), copper (Cu), and zinc (Zn) were found in lower concentrations, while molybdenum (Mo) and silver (Ag) occurred at trace levels. The positive correlation of Ag, Mo, and some bacterial genera, like *Microcystis* and *Mycobacterium*, indicates possible interactions via biosorption or enzymatic mechanisms [47]. These elements could serve as co-factors for microbial enzymes involved in metal detoxification and nutrient cycling. The statistical analyses consistently demonstrated significant spatial variation in elemental concentrations across sites. Both parametric (one way ANOVA) and non-parametric (Kruskal Wallis) tests confirmed that differences were not random but reflect genuine heterogeneity in elemental composition, likely driven by site-specific environmental or anthropogenic factors.

The biochemical oxygen demand (BOD) and chemical oxygen demand (COD) concentrations in the seven analyzed samples (KES0–KES6) indicate varying degrees of organic pollution in the studied water bodies. The BOD values ranged from 5.6 ppm (KES0) to 22.8 ppm (KES4), while COD values ranged from 21.9 ppm (KES0) to 218.9 ppm (KES4). The significantly high BOD and COD concentrations in KES4 suggest a high level of organic contamination, likely due to anthropogenic influences, which was low when compared to other sites. Comparisons with other studies indicate similar trends in water bodies affected by human activities, though the sources and extent of pollution may vary. For instance, Dodi Tal in the Garhwal Himalaya has shown BOD levels correlating with phytoplankton density, suggesting nutrient enrichment leading to eutrophication [57]. Similarly, Loktak Lake in Manipur has recorded BOD values ranging from 0.99 to 4.19 ppm, with higher values linked to domestic waste inputs [11]. In contrast, the elevated BOD and COD levels observed in our study are largely attributed to organic inputs from regular fish feeding practices, rather than municipal or industrial discharge. This distinction underscores the need to consider local anthropogenic influences when assessing water quality and eutrophication in high-altitude freshwater systems [11]. Similarly, a study on lake pollution in China showed that COD levels above 150 ppm were associated with algal bloom formation and hypoxic conditions, further affecting aquatic ecosystems [35]. The strong degradation gradient observed in Khecheopalri Lake from pristine conditions at KES0 (turbidity 2.0 NTU; DO 9.0 ppm; PLI = 0.85) to severe contamination at KES4 (turbidity 13.0 NTU; DO 4.8 ppm; PI_Fe ≈115, PI_Mn ≈56, PI_Cr ≈223; PLI = 2.10) parallels similar findings in Aritar Lake, where spatial and seasonal nutrient enrichment (NI up to 7.31 pre-monsoon) and Class II organic pollution (OPI) indicate progressing eutrophication risk [2]. Together, these studies underscore how mountain lakes in the Himalayas can rapidly shift from oligotrophic to degraded states under anthropogenic and basin-level pressures. The analyses clearly demonstrate significant spatial heterogeneity in both physicochemical parameters and organic load across Khecheopalri Lake. Strong site effects (ICC = 0.734; $\omega^2 = 0.702$) indicate that environmental conditions are strongly structured by location, with KES3 and KES4 representing critical hotspots of degradation. Similarly, BOD and COD showed moderate but significant site-related variation (ICC = 0.463; $\omega^2 = 0.302$), reflecting localized organic pollution inputs. The consistency between parametric and non-parametric tests, despite variance heterogeneity, reinforces the robustness of these findings and highlights spatially uneven pollution pressures within the lake.

Shotgun metagenomic sequencing analysis of the freshwater contaminated lake revealed interesting information on the microbial community structure and its relationship with other environmental factors. The results point towards a very high bacterial dominance (98.85%), with lower percentages of eukaryotes, viruses, archaea, protists, and plantae. The overall microbial diversity disclosed the occurrence of six principal phyla where Pseudomonadota were found most dominant in samples KES0, KES1, KES2, KES5, and KES6, while Cyanobacteria dominated in KES3 and KES4. The dominance of Pseudomonadota in eutrophic or polluted freshwater bodies has been reported in multiple studies, supporting their adaptability to nutrient-rich environments [31]. The high abundance of *Microcystis* in KES3 and KES4 aligns with previous studies that have linked *Microcystis* blooms to increased nutrient loading in freshwater ecosystems [58]. The presence of *Microcystis novacekii* in KES4 further highlights the potential for cyanobacterial blooms under favorable conditions such as high nitrogen and phosphorus levels [27]. At the family level, Enterobacteriaceae was found in high abundance in KES0, KES1, KES2, KES5, and KES6. This is consistent with other studies that have associated Enterobacteriaceae with fecal contamination possibly of birds and animal origin [33]. In contrast, Microcystaceae was the dominant family in KES3 and KES4, which is in agreement with studies by Maurer et al. [38] reporting Microcystaceae as the primary family involved in cyanobacterial blooms in urban lakes. The observed shifts in microbial composition suggest that anthropogenic pressures favor opportunistic and bloom-forming taxa while suppressing ecologically important genera. Such alterations in community structure may disrupt nutrient cycling, reduce water quality, and increase the risk of harmful algal blooms, underscoring the ecological consequences of human-induced disturbances.

A culture-dependent analysis identified 18 bacterial isolates from the genera *Limnohabitans, Mycolicibacterium, Microcystis,* and *Micromonospora*. *Limnohabitans planktonicus, Limnohabitans curvas,* and *Limnohabitans parvus* were isolated from KES0, KES1, KES2, KES5, and KES6, where algal blooms were less prominent on the surface. Previously, *Limnohabitans planktonicus* sp. nov. and *Limnohabitans parvus* sp. nov. were also isolated from the meso-eutrophic freshwater Římov Reservoir in the Czech Republic [26]. *Microcystis aeruginosa* was found in KES3 and KES4 of Khecheopalri Lake, which as observed in Chilika Lake, India's largest brackish water lagoon [44, 60]. The study identified factors such as elevated water temperatures (25.31–32.48 °C), increased dissolved inorganic nitrogen, strong phosphorus limitation, higher water transparency, and low salinity as contributors to the bloom's rapid onset. These conditions facilitated the proliferation of *M. aeruginosa*, which dominated the phytoplankton community during the bloom period [60]. In the United States, a notable incident occurred in Lake Erie, where a toxic algal bloom contaminated Toledo's water supply. This event prompted researchers to investigate the health effects of such blooms, focusing on how microcystin exposure from *Microcystis aeruginosa* affected individuals with conditions like asthma and other respiratory diseases. Their studies revealed that exposure to aerosolized toxins could exacerbate inflammation, particularly in asthma patients [58]. The concordance between taxa identified via culture-based techniques and those recovered through shotgun metagenomic sequencing enhances confidence in the taxonomic assignments and the overall microbial community profile. This cross-validation approach strengthens the ecological interpretation of the data, confirming the presence of key genera such as *Microcystis* and *Limnohabitans* and supporting their ecological associations with nutrient levels and bloom formation.

Highest values of the diversity indices (Shannon, Simpson, and Evenness) in KES5 indicate a presumably stable and rich microbial community for that sample. The highest value of the Chao-1 richness index occurred in KES1, pointing to the most species being represented uniquely. Canonical correlation analysis (CCA) on the genus level indicated significant relationships between water quality parameters like BOD, COD, and microbiological composition. The positive correlation of *Microcystis, Mycobacterium, Klebsiella*, and *Polynucleobacter* with BOD and COD implies their possible use as bioindicators of organic pollution. *Mycobacterium* and *Klebsiella* have also been associated with closed water body causing eutrophication [18, 49]. In addition, elemental analysis also identified a strong link between *Microcystis, Mycobacterium, Bradyrhizobium*, and trace metals Ag and Mo. *Microcystis* correlation with trace metals might suggest biosorption or enzymatic interactions as already reported in earlier research on cyanobacterial metal accumulation [24]. In the same vein, *Bradyrhizobium* also needs molybdenum for fixing nitrogen, and the result was as found earlier through previous studies demonstrating the significance of molybdenum in nitrogenase activity [34].

Metagenomic analysis showed that 72% of predicted ORFs were annotated to KEGG pathways, with metabolism being the most dominant category (71.64%), followed by genetic information processing (6.95%) and environmental information processing (5.96%). Carbohydrate metabolism (15.04%) and amino acid metabolism (7.09%) were the most represented level-2 sub-pathways, indicating active nutrient cycling. The functional shifts indicate that anthropogenic influence drives microbial communities toward stress-response and pollutant-degrading capacities while diminishing core cellular processes. The enrichment of pathways linked to infectious diseases and cancer further highlights potential public health risks and underscores the ecological consequences of human-driven disturbances.

Notably, xenobiotic degradation pathways accounted for 3.14% of mapped functions, reflecting the microbial community's capacity to transform environmental pollutants. Genes encoding enzymes such as *xylB*, *pchF*, and *clcD* (toluene degradation), *thlA*, *phaA*, and *mmgA* (benzoate degradation), and *mhpD*, *mhpE* (xylene degradation) were detected, revealing potential for aromatic hydrocarbon metabolism [40, 56]. Microbial taxa showed specific functional correlations: *Limnohabitans planktonicus* (KES0–KES2) with purine metabolism (KO00230), *Candidatus Planktophila dulcis* (KES3) with quorum sensing (KO02024), *Microcystis aeruginosa* (KES4) with xylene degradation (KO00622), and *Limnohabitans radicicola* (KES6) with pyruvate metabolism (KO00620). These associations suggest an ecologically responsive microbial structure capable of adapting to nutrient and pollutant inputs. The presence of xenobiotic degradation genes underscores the potential role of lake microbiota in mitigating anthropogenic impacts and maintaining ecological balance [24, 29]. Ongoing monitoring and integration of functional and taxonomic data can provide a clearer picture of ecosystem resilience.

The findings from Khecheopalri Lake indicate significant microbial shifts. Elevated phosphorus, iron, and potassium levels suggest nutrient enrichment, likely from human activity. High BOD and COD values in KES4 point to severe organic contamination, promoting the growth of cyanobacterial species like *Microcystis aeruginosa*, known for toxic blooms. Potential sources of this organic contamination may include food offerings and rituals performed by visitors, runoff from nearby settlements, decaying plant material, and uneaten fish feed from aquaculture practices. These inputs can collectively increase nutrient loads, fuelling microbial imbalances and eutrophication. The detection of genes involved in xenobiotic degradation pathways suggests that the microbial community in Khecheopalri Lake possesses the metabolic potential to break down harmful aromatic compounds such as toluene, xylene, and benzoate. Enzymes like *xylB, pchF*, and *clcD* point to active toluene metabolism, while the presence of *thlA, phaA,* and *mmgA* supports benzoate degradation. The identification of *mhpD* and *mhpE* further indicates xylene breakdown potential. Despite the pristine habitat, the occurrence of xenobiotic and hydrocarbon degradation genes could be the part of the inherent microbial arsenal. These findings underscore the diversity and ecological significance of these microbes in maintaining lake health through biogeochemical and natural detoxification processes.

Conclusion

Khecheopalri Lake, a sacred and ecologically sensitive high-altitude freshwater system in Sikkim, is showing signs of environmental stress, as indicated by elevated levels of phosphorus, nitrogen, and trace metals. This is further evidenced by increased biological oxygen demand (BOD) and chemical oxygen demand (COD) at specific sites, likely due to the accumulation of organic matter from fish feeding practices and the lake's semi-closed hydrological system, which limits natural flushing and dilution. The detection of cyanobacterial blooms such as *Microcystis aeruginosa* highlights the ongoing eutrophication process. Shotgun metagenomic analyses supported by the culture-dependent approach revealed a diverse microbial community, including potentially pathogenic taxa. This study provides the first comprehensive insight into the microbial assemblages and their functional attributes in Khecheopalri Lake, offering early indications of ecological perturbations. The results form a critical baseline for long-term monitoring and suggest that informed conservation strategies will be essential to maintain the lake's ecological integrity and sacredness. However, the study is limited by its single-season sampling and lack of temporal replication, which restricts broader generalizations about seasonal or long-term trends. Future studies should therefore incorporate multi-seasonal and longitudinal monitoring, integrate functional assays with metagenomics, and assess ecosystem-level impacts to better guide sustainable lake management and conservation strategies.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12866-025-04450-1.

Supplementary Material 1.

Supplementary Material 2.

Acknowledgements

The authors heartily thank and express their most sincere gratitude to the Sikkim State Biodiversity Board & Forest, Environment, and Wildlife Management Department, Govt. of Sikkim, Sikkim Police, and North Sikkim administration, for granting research permission, kind cooperation, and support during the fieldwork. No extramural fund is being provided for APC of the manuscript.

Authors' contributions

R. D conceptualise and completed the experiments, data analysis and wrote the manuscript. B. T. sampling, data analysis and wrote the manuscript. R. K. data analysis, revised and edited the manuscript. N. T. data analysis, revised and edited the manuscript. B. T.- conceptualise the experiments, data analysis, supervised, and edited the manuscript.All authors reviewed the manuscript.

Funding

The study was conducted without financial assistance from public, commercial, or non-profit organizations.

Data availability

Data has been submitted to NCBI with BioProject number for sample KES0 (PRJNA1281409), KES1 (PRJNA1281371), KES2 (PRJNA1281371), KES3 (PRJNA1281549), KES4 (PRJNA1281409), KES5 (PRJNA1281549) and KES6 (PRJNA1281549). The accession numbers of the isolated strains identified through Sanger sequencing has been mentioned in the Supplementary Fig. 1.

Declarations

Competing interest

The authors declare no competing interests.

Received: 21 July 2025 / Accepted: 3 October 2025
Published online: 31 October 2025

References

1. Ashade AO, Obayori OS, Salam LB, Fashola MO, Nwaokorie FO. Effects of anthropogenic activities on the microbial community diversity of Ologe Lagoon sediment in Lagos State, Nigeria. Environ Monit Assess. 2024. https://doi.org/10.1007/s10661-024-13025-y.
2. Auddy N, Sinha R, Das Bhattacharya S, Pobi KK, Dutta S, Nayek S. Application of water quality indices and geostatistical methods for analysing mountain lakes in relation to anthropogenic influences and catchment features: a case study in East Sikkim, India. Water Pract Technol. 2023;18(11):2638–52. https://doi.org/10.2166/wpt.2023.172.
3. Ayala-Muñoz D, Macalady JL, Sánchez-España J, Falagán C, Couradeau E, Burgos WD. Microbial carbon, sulfur, iron, and nitrogen cycling linked to the potential remediation of a meromictic acidic pit lake. ISME J. 2022;16(12):2666–79. https://doi.org/10.1038/s41396-022-01320-w.
4. Bolger AM, Lohse M, Usadel B. Trimmomatic: A flexible trimmer for Illumina sequence data. Bioinformatics. 2014;30(15):2114–20. https://doi.org/10.1093/bioinformatics/btu170.
5. Brandt J & Albertsen M. Investigation of detection limits and the influence of DNA extraction and primer choice on the observed microbial communities in drinking water samples using 16S rRNA gene amplicon sequencing. Front Microbiol. 2018; 9(SEP). https://doi.org/10.3389/fmicb.2018.02140
6. Cantarero A, López MB, Mahía J, Maestro MA, Paz A. Determination of total and dissolved phosphorus in agricultural runoff samples by inductively coupled plasma mass spectrometry. Commun Soil Sci Plant Anal. 2002;33(15–18):3431–6. https://doi.org/10.1081/CSS-120014536.
7. Chen F, Jiang F, Ma J, Alghamdi MA, Zhu Y, Yong JWH. Intersecting planetary health: exploring the impacts of environmental stressors on wildlife and human health. Ecotoxicol Environ Saf. 2024. https://doi.org/10.1016/j.ecoenv.2024.116848.
8. Chen X, Sheng Y, Wang G, Zhou P, Liao F, Mao H, et al. Spatiotemporal successions of N, S, C, Fe, and As cycling genes in groundwater of a wetland ecosystem: enhanced heterogeneity in wet season. Water Res. 2024. https://doi.org/10.1016/j.watres.2024.121105.
9. Cui H, Ou Y, Wang L, Liang A, Yan B, Li Y. Dynamic changes in microbial communities and nutrient stoichiometry associated with soil aggregate structure in restored wetlands. CATENA. 2021. https://doi.org/10.1016/j.catena.2020.104984.
10. Dai H, Zhang Y, Fang W, Liu J, Hong J, Zou C, et al. Microbial community structural response to variations in physicochemical features of different aquifers. Front Microbiol. 2023. https://doi.org/10.3389/fmicb.2023.1025964.
11. Das Kangabam R, Bhoominathan SD, Kanagaraj S, Govindaraju M. Development of a water quality index (WQI) for the Loktak Lake in India. Appl Water Sci. 2017;7(6):2907–18. https://doi.org/10.1007/s13201-017-0579-4.
12. Das S, Tamang JP. Changes in microbial communities and their predictive functionalities during fermentation of toddy, an alcoholic beverage of India. Microbiol Res. 2021. https://doi.org/10.1016/j.micres.2021.126769.
13. Dong X, Strous M. An integrated pipeline for annotation and visualization of metagenomic contigs. Front Genet. 2019. https://doi.org/10.3389/fgene.2019.00999.
14. Evershed S & Fish A. In The Middle of The Lotus: Khecheopalri Lake, A Contested Sacred Land in The Eastern Himalaya of Sikkim. 2006;14(17):5165.http://www.indiatravelogue.com/dest/sik/sikkim2.html.

15. Freitas TAK, Li PE, Scholz MB & Chain PSG. Accurate read-based metagenome characterization using a hierarchical suite of unique signatures. Nucleic Acids Res. 2015; 43(10). https://doi.org/10.1093/nar/gkv180
16. Gao H, Span I. The diversity and applications of microbial iron metabolism and iron-containing proteins. Commun Biol. 2025;8(1):177. https://doi.org/10.1038/s42003-024-07373-1.
17. Hammer DAT, Ryan PD, Hammer Ø & Harper DAT. Past: Paleontological Statistics Software Package for Education and Data Analysis. In Palaeontologia Electronica. 2001;4(1):49. http://palaeo-electronica.org/2001_1/past/issue1_01.htm.
18. Henriot CP, Martak D, Genet S, Bornette G, Hocquet D. Origin, fluxes, and reservoirs of *Escherichia coli*, *Klebsiella pneumoniae*, and *Pseudomonas aeruginosa* in aquatic ecosystems of a French floodplain. Sci Total Environ. 2022. https://doi.org/10.1016/j.scitotenv.2022.155353.
19. Howarth RW, Chan F, Swaney DP, Marino RM, Hayn M. Role of external inputs of nutrients to aquatic ecosystems in determining prevalence of nitrogen vs. phosphorus limitation of net primary productivity. Biogeochemistry. 2021;154(2):293–306. https://doi.org/10.1007/s10533-021-00765-z.
20. Hur J, Lee BM, Lee TH, Park DH. Estimation of biological oxygen demand and chemical oxygen demand for combined sewer systems using synchronous fluorescence spectra. Sensors. 2010;10(4):2460–71. https://doi.org/10.3390/s100402460.
21. Hyatt D, Chen G-L, Locascio PF, Land ML, Larimer FW, Hauser LJ. Prodigal: prokaryotic gene recognition and translation initiation site identification. BMC Bioinformatics. 2010. https://doi.org/10.1186/1471-2105-11-119.
22. Jain A, Rai SC, Pal J & Sharma E. Hydrology and nutrient dynamics of a sacred lake in Sikkim Himalaya. In Hydrobiologia. 1999 (Vol. 416).
23. Jain A, Rai SC, Sharma E. Hydro-ecological analysis of a sacred lake watershed system in relation to land-use/cover change from Sikkim Himalaya. CATENA. 2000;40(3):263–78. https://doi.org/10.1016/S0341-8162(00)00086-2.
24. Kalita N, Baruah PP, Elsevier B.V. Cyanobacteria as a potent platform for heavy metals biosorption: uptake, responses and removal mechanisms. J Hazardous Mat Adv. 2023. https://doi.org/10.1016/j.hazadv.2023.100349.
25. Kanehisa M, Goto S. KEGG: Kyoto encyclopedia of genes and genomes. Nucleic Acids Res. 2000;28(1):27–30.
26. Kasalický V, Jezbera J, Šimek K, Hahn MW. *Limnohabitans* planktonicus sp. nov. and *Limnohabitans* parvus sp. nov., planktonic betaproteobacteria isolated from a freshwater reservoir, and emended description of the genus *Limnohabitans*. Int J Syst Evol Microbiol. 2010;60(12):2710–4. https://doi.org/10.1099/ijs.0.018952-0.
27. Kim K, Mun H, Shin H, Park S, Yu C, Lee J, et al. Nitrogen stimulates *Microcystis*-dominated blooms more than phosphorus in river conditions that favor non-nitrogen-fixing genera. Environ Sci Technol. 2020;54(12):7185–93. https://doi.org/10.1021/acs.est.9b07528.
28. Kramer BJ, Turk-Kubo K, Zehr JP, Gobler CJ. Intensification of harmful cyanobacterial blooms in a eutrophic, temperate lake caused by nitrogen, temperature, and CO2. Sci Total Environ. 2024. https://doi.org/10.1016/j.scitotenv.2024.169885.
29. Kugarajah V, Nisha KN, Jayakumar R, Sahabudeen S, Ramakrishnan P, Mohamed SB, et al. Significance of microbial genome in environmental remediation. Microbiol Res. 2023. https://doi.org/10.1016/j.micres.2023.127360.
30. Laslett D, Canback B. ARAGORN, a program to detect tRNA genes and tmRNA genes in nucleotide sequences. Nucleic Acids Res. 2004;32(1):11–6. https://doi.org/10.1093/nar/gkh152.
31. Lennert KJ, Borsodi AK, Anda D, Krett G, Kós PB, Engloner AI. The effect of urbanization on planktonic and biofilm bacterial communities in different water bodies of the Danube River in Hungary. Sci Rep. 2024;14(1):23881. https://doi.org/10.1038/s41598-024-75863-7.
32. Letunic I, Bork P. Interactive tree of life (iTOL) v5: an online tool for phylogenetic tree display and annotation. Nucleic Acids Res. 2021;49(W1):W293–6. https://doi.org/10.1093/nar/gkab301.
33. Lew S, Glińska-Lewczuk K, Burandt P, Grzybowski M, Obolewski K. Fecal bacteria in coastal lakes: an anthropogenic contamination or natural element of microbial diversity? Ecol Indic. 2023. https://doi.org/10.1016/j.ecolind.2023.110370.
34. Li M, Zhang P, Guo Z, Cao W, Gao L, Li Y, et al. Molybdenum nanofertilizer boosts biological nitrogen fixation and yield of soybean through delaying nodule senescence and nutrition enhancement. ACS Nano. 2023;17(15):14761–74. https://doi.org/10.1021/acsnano.3c02783.
35. Li S, Liu C, Sun P, Ni T. Response of cyanobacterial bloom risk to nitrogen and phosphorus concentrations in large shallow lakes determined through geographical detector: a case study of Taihu Lake, China. Sci Total Environ. 2022. https://doi.org/10.1016/j.scitotenv.2021.151617.
36. Lin H, Peddada SD. Analysis of compositions of microbiomes with bias correction. Nat Commun. 2020. https://doi.org/10.1038/s41467-020-17041-7.
37. Martino C, Morton JT, Marotz CA, Thompson LR, Tripathi A, Knight R, et al. A novel sparse compositional technique reveals microbial perturbations. MSyst. 2019. https://doi.org/10.1128/msystems.00016-19.
38. Maurer JA, Kim AM, Oblie N, Hefferan S, Xie H, Slitt A, et al. Temporal Dynamics of Cyanobacterial Bloom Community Composition and Toxin Production from Urban Lakes. 2024. https://doi.org/10.1101/2024.02.07.579333.
39. Metsalu T, Vilo J. ClustVis: a web tool for visualizing clustering of multivariate data using principal component analysis and heatmap. Nucleic Acids Res. 2015;43(W1):W566–70. https://doi.org/10.1093/nar/gkv468.
40. Miglani R, Parveen N, Kumar A, Ansari MA, Khanna S, Rawat G, et al. Degradation of xenobiotic pollutants: an environmentally sustainable approach. Metabolites. 2022. https://doi.org/10.3390/metabo12090818.
41. Najar IN, Sherpa MT, Das S, Das S, Thakur N. Microbial ecology of two hot springs of Sikkim: predominate population and geochemistry. Sci Total Environ. 2018;637:730–45. https://doi.org/10.1016/j.scitotenv.2018.05.037.
42. Najar IN, Sherpa MT, Das S, Verma K, Dubey VK, Thakur N. Geobacillus yumthangensis sp. nov., a thermophilic bacterium isolated from a north-east indian hot spring. Int J Syst Evol Microbiol. 2018;68(11):3430–4. https://doi.org/10.1099/ijsem.0.003002.
43. Nurk S, Meleshko D, Korobeynikov A, Pevzner PA. Metaspades: a new versatile metagenomic assembler. Genome Res. 2017;27(5):824–34. https://doi.org/10.1101/gr.213959.116.
44. Olalekan Ashade A, Sunday Obayori O, Omotola Fashola M, Obiageri Nwaokorie F. Assessments of the impacts of chemical and hydrocarbon pollution on Ologe Lagoon water and its prokaryotic community structure. Aquat Ecol. 2024. https://doi.org/10.1007/s10452-024-10102-9.
45. O'Neill AR, Chhetri PK, Chhetri B, Rana SK. Establishing ecological baselines around a temperate Himalayan peatland. Wetl Ecol Manage. 2020;28(Issue 2):375–88. https://doi.org/10.1007/s11273-020-09710-7.
46. Packa V, Maedler S, Howell T, Bostan V, Diep N, Tooley R, et al. Unbiased measurement of phosphate and phosphorus speciation in surface waters. Environ Sci Technol. 2019;53(2):820–8. https://doi.org/10.1021/acs.est.8b05089.
47. Pande V, Pandey SC, Sati D, Bhatt P, Samant M. Microbial interventions in bioremediation of heavy metal contaminants in agroecosystem. Front Microbiol. 2022. https://doi.org/10.3389/fmicb.2022.824084.
48. Patel RK, Jain M. NGS QC toolkit: a toolkit for quality control of next generation sequencing data. PLoS One. 2012. https://doi.org/10.1371/journal.pone.0030619.
49. Primm TP, Lucero CA, Falkinham JO. Health impacts of environmental mycobacteria. Clin Microbiol Rev. 2004;17(1):98–106. https://doi.org/10.1128/CMR.17.1.98-106.2004.
50. Priya RP & Malhotra A. Biodiversity Strategy and Action Plan (BSAP) of Sikkim and the Resource Mobilisation Strategy for implementing the BSAP with focus on Khangchendzonga-Upper Teesta Valley. 2021.
51. Raina KV. Geological mapping in the western part of Sikkim. 1966. http://www.indiatravelogue.com/dest/sik/sikkim2.html
52. Read EK, Ivancic M, Hanson P, Cade-Menun BJ, McMahon KD. Phosphorus speciation in a eutrophic lake by ^{31}P NMR spectroscopy. Water Res. 2014;62:229–40. https://doi.org/10.1016/j.watres.2014.06.005.
53. Reitzel K, Ahlgren J, Gogoll A, Jensen HS, Rydin E. Characterization of phosphorus in sequential extracts from lake sediments using ^{31}P nuclear magnetic resonance spectroscopy. Can J Fish Aquat Sci. 2006;63(8):1686–99. https://doi.org/10.1139/F06-070.
54. Saitou N & Nei M. The Neighbor-joining Method: A New Method for Reconstructing Phylogenetic Trees'.1987.
55. Seemann T. Prokka: rapid prokaryotic genome annotation. Bioinformatics. 2014;30(14):2068–9. https://doi.org/10.1093/bioinformatics/btu153.
56. Seo JS, Keum YS & Li QX. (2009). Bacterial degradation of aromatic compounds. Int J Environment Res Public Health. 2009; 6(1):278–309. MDPI. https://doi.org/10.3390/ijerph6010278
57. Singh S, Nandan H & Sharma RC. Zooplankton diversity and potential indicator species for assessment water quality of high altitude wetland, Dodi Tal of Garhwal Himalaya, India. Academia Arena. 2020; 5(12). https://doi.org/10.7537/marsaaj120520.01
58. Steffen, Morgan M. and Davis, Timothy W. and McKay, R. Michael L. and Bullerjahn, George S. and Krausfeldt, Lauren E. and Stough, Joshua M.A. and Neitzey, Michelle L. and Gilbert, Naomi E. and Boyer, Gregory L. and

Johengen, Thomas H. and Gossiaux, Duane C. and Burtner, Ashley M. and Palladino, Danna and Rowe, Mark D. and Dick, Gregory J. and Meyer, Kevin A. and Levy, Shawn and Boone, Braden E. and Stumpf, Richard P. and Wynne, Timothy T. and Zimba, Paul V. and Gutierrez, Danielle and Wilhelm, Steven W. Ecophysiological Examination of the Lake Erie Microcystis Bloom in 2014: Linkages between Biology and the Water Supply Shutdown of Toledo, OH. Environ. Sci. Tachnol. 2017;51(12):6745-7655. https://doi.org/10.1021/acs.est.7b00856.http://pubs.acs.org.

59. Tamura K, Nei M, Kumar S. Prospects for inferring very large phylogenies by using the neighbor-joining method. Proc Natl Acad Sci U S A. 2004. https://doi.org/10.1073/pnas.0404206101.
60. Tarafdar L, Mohapatra M, Muduli PR, Kumar A, Mishra DR, Rastogi G. Co-occurrence patterns and environmental factors associated with rapid onset of *Microcystis aeruginosa* bloom in a tropical coastal lagoon. J Environ Manage. 2023. https://doi.org/10.1016/j.jenvman.2022.116580.
61. Thompson JD, Higgins DG, Gibson TJ. CLUSTAL W: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice. Nucleic Acids Res. 1994. https://doi.org/10.1093/nar/22.22.4673.
62. Uprety IM, & Sharma G. Cultural Attributes, Economic Valuation and Community Conservation in Holy Khecheopalri Lake of Sikkim in the Eastern Himalaya. 2012.
63. Waldemer C, Lechtenfeld OJ, Gao S, Koschorreck M, Herzsprung P. Anaerobic degradation of excess protein-rich fish feed drives CH4 ebullition in a freshwater aquaculture pond. Sci Total Environ. 2024. https://doi.org/10.1016/j.scitotenv.2024.176514.

Publisher's Note